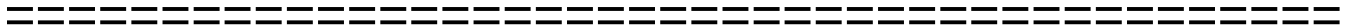




Centre for
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CMInDS
Indian Institute of Technology Bombay

Programming for Machine Learning and Data Science



Course Project Description Document

Automated Detection of Teams of Players Using IPL Jersey Identification

Overview:

- This project carries 40 marks (20% weightage)
- It must be done in groups of max 4 members.
- **Submission deadline:** June 06, 2026, 23:55 Hrs
- Evaluation Mode: Based on the presentation, video, and submitted artifacts (dataset, code, weights, CSV outputs).

Problem Description:

The objective of this project is to train and test machine learning models with the following capabilities:

- The model should systematically examine regions within an image to determine whether they contain players and classify each detected player's team according to the IPL franchise they belong to, based on visual cues such as jersey color, patterns, and logos.
- Note that only the presence of players, and their team affiliation is expected. You are NOT required to identify their names!

For example, given an image from the 18th IPL season (2025) that includes five players from different teams, the system should be able to:

1. Detect the presence of all individual players in the image.
2. Accurately count the number of players.
3. Assign each player to the correct IPL franchise.

To accomplish this task, you are required to build your own dataset. This involves collecting cricket images from reliable online sources, preprocessing them, and annotating them with relevant labels such as player bounding boxes and corresponding team identities. The final model

should demonstrate the ability to generalize across diverse match scenarios, lighting conditions, and player appearances.

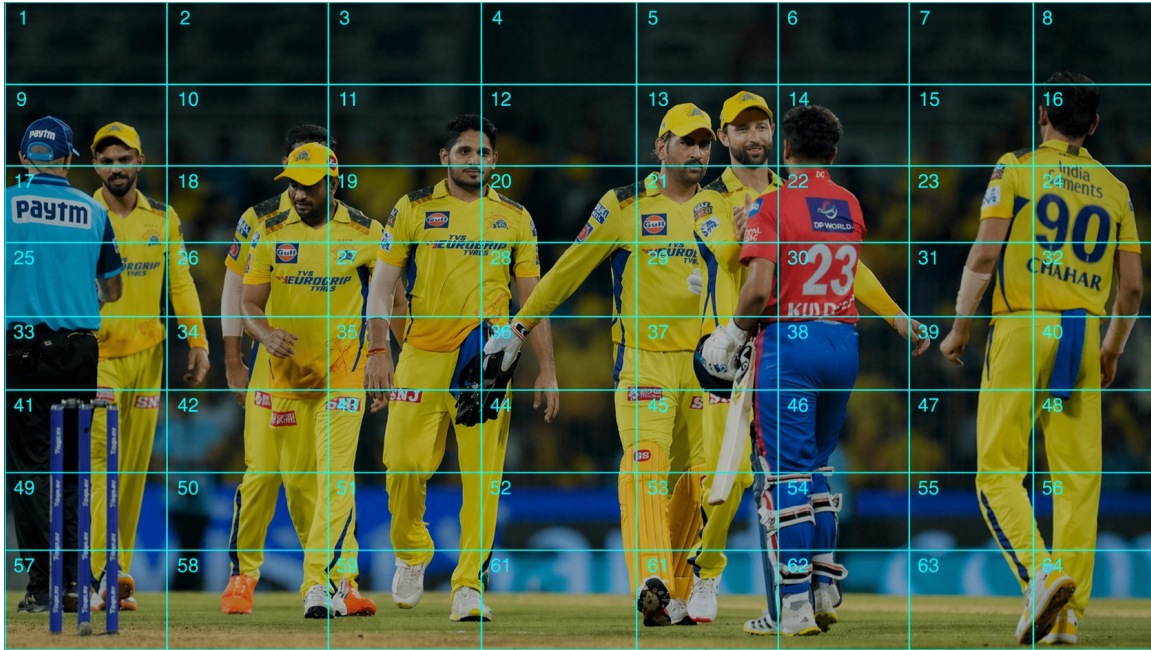
Dataset:

- The data set (collection of images) needs to be created. The data set should ensure that each franchise is adequately represented. It is recommended that there should be at least 100 instance of each franchise.
- **Teams are strongly encouraged to collaborate on the creation of a common data set to ensure a significantly large and balanced 'labelled' data set, and label it.**
- Images should contain:
 - Multiple players from different IPL franchises in a single frame.
 - Players wearing distinct team jerseys (various colors, designs, and logos).
 - Variations in player poses, orientations, and partial/full body visibility.
 - Different match conditions such as lighting, camera angles, crowd backgrounds, and occlusions.
 - Single player and multi-player scenes to help with both detection and counting tasks.
 - No-player images (e.g., empty pitch, crowd, or background scenes) to improve model generalization.
- Dataset requirements:
 - All images must have a 4:3 aspect ratio and be resized to 800 x 600 pixels.
 - Note: It is acceptable to downsize higher resolution images. Images with a resolution lower than 800x600 should NOT be scaled up / used
 - Include a short README file describing image sources.

Modeling Task:

- Divide each 800 x 600 image into an 8×8 grid (64 cells), as shown in the image below.

- Predict for each grid cell:
 - 0 :No team
 - 1 :Chennai Super Kings (CSK)
 - 2 :Delhi Capitals (DC)
 - 3 :Gujarat Titans (GT)
 - 4 :Kolkata Knight Riders (KKR)
 - 5 :Lucknow Super Giants (LSG)
 - 6 :Mumbai Indians (MI)
 - 7 :Punjab Kings (PBKS)
 - 8 :Rajasthan Royals (RR)
 - 9 :Royal Challengers Bengaluru (RCB)
 - 10 :Sunrisers Hyderabad (SRH)
 - Note: In case multiple objects are present in a cell, the classifier should detect any ONE of them.
- Any **hand-crafted feature engineering techniques for images** can be used. **This implies that methods like CNNs, or equivalents, that automatically create features from images, SHOULD NOT be used.**
- Save the trained model as model_<teamname>.pkl.
- Run your final (best) model on both, the train and test, datasets and capture the outcome in a CSV file having the following format:
 - Image File Name, Train Or Test, c01, c02, ..., c64
 - The value in each column should be 0-10 depending on the prediction, as enumerated above.



For example, the entry in the CSV file for the above image will look as follows

- Column c29 should be marked as 1 since it contains the team CSK.
- Columns c22, c30, and c62 should be marked as 2 since they contain the team DC.
- Columns c1-c8 should be marked as 0, indicating no object.

Deliverables:

<u>Deliverable</u>	<u>Description</u>
Dataset	Count and size as outlined earlier in this document (with folder structure and README.txt explaining sources).
Source Code	All scripts/notebooks used for data processing, feature creation, model training, and inference.
Performance Metrics	On train/test data
Model Weights	Trained pickle file (.pkl).

Pipeline Code	To read the pickle file, accept a 'test' instance and create the required output
Predictions CSV	Output file as specified above.
Presentation	Well-structured slide deck summarizing approach, methodology, metrics, and results. See Presentation Guidelines below for more information
Video (about 5 minutes)	Overview of the entire workflow – data, features, model, results, challenges, and key learnings.

Presentation Guidelines

- There is no limit on the number of slides in the presentation. The presentation should document all aspects of the project, but precisely and in a compact manner.
- The first/title slide should contain the roll numbers and names of all the team members.
- Following the title slide there should be maximum 3 slides containing an executive summary,
- As mentioned above, the remaining slides should capture all aspects of the project: problem explanation; details of all the paths and steps taken; metrics; observations, analysis, conclusions; challenges, and learnings.
 - Note: Include all aspects of the project, including paths taken that resulted in dead-ends / failures / non-optimal results, etc. That is, it should document the entire efforts of the team.
- The video time should not exceed 5 minutes, and if it does by more than 15 seconds there will be a penalty.

Evaluation criteria

Criteria	Marks
Problem detailing, solution approach, completeness of solution, and Data Science steps	15

followed	
Quality of results on the train/test data and also on hold-out data (i.e, unseen data set used for evaluating the models); observations, analysis, conclusions	15
Documentation: Completeness and quality of the presentation and video	10